Find the Laplace transform of the following:

1. $e^{2t}\cos^2 t$ ans: $\dfrac{s^2-4s+6}{(s-2)(s^2-4s+8)}$

2. $t^2e^{-3t}\sin 2t$ ans: $\dfrac{4(3s^2+18s+23}{(s^2+6s+13)^3}$

3. $\sinh(3t)\cos^2 t$ ans: $\dfrac{3}{2}\left[\dfrac{1}{s^2-9}+\dfrac{s^2-13}{s^4-10s^2+169}\right]$

4. $t^2e^{-3t}\sin(2t)$ ans: $\dfrac{4(3s^2+18s+23)}{(s^2+6s+13)^3}$

5. $t\cos^3 t$ ans: $\dfrac{1}{4}\left[\dfrac{3(s^2-1)}{(s^2+1)^2}+\dfrac{s^2-9}{(s^2+9)^2}\right]$